

MD. Tanvir Shakil

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 MD Tanvir Shakil

Nilphamari, Uttora EPZ, Bangladesh

OBJECTIVE

Electronics Engineer with expertise in multi-layer PCB design, embedded systems, IoT/AIoT architectures, and embedded AI. Proven experience leading end-to-end hardware development, from system architecture and component selection to prototyping, validation, and deployment. Focused on advancing e-mobility, EV charging infrastructure, and smart power-electronic solutions through robust design, intelligent control, and high-reliability engineering practices.

RESEARCH EXPERIENCE

- **Bangladesh Agricultural University-Research and Innovation Centre (BAU-RIC)**  Nov 2024 – Nov 2025
Research Associate – IoT Engineer
 - An Artificial Intelligence-Based Harness Tag for Estrus Detection in Cattle (PCB Design, Firmware, and System Design)
- **Bangladesh Space Research and Remote Sensing Organization (SPARRSO)**  FY 2023 – FY 2025
Junior Research Fellow Dhaka, Bangladesh
 - Design and Manufacturing of a 1U CubeSat with Optimized Chassis and 3D Printed Carbon Fiber Composite. (PCB Design and Firmware)
 - Development of an Engineering Model CubeSat and Prototype Solid Propulsion System for Space Research. (PCB Design and Firmware)

INDUSTRY EXPERIENCE

- **THT-Space Electrical Company Ltd.**  Present
Embedded Software Engineer (AI) Nilphamari, Bangladesh
 - Developing AI-driven embedded solutions for smart home electronics and intelligent automation systems.
 - Designing and integrating AIoT-enabled devices with wireless connectivity (Wi-Fi, Bluetooth, and MQTT) for real-time monitoring and control.
 - Leading cross-functional teams in firmware development, AIoT hardware-software integration, and system optimization.
 - Implementing energy-efficient and user-centric features to enhance the reliability, safety, and scalability of smart systems.
- **Palki Motors Ltd.**  Dhaka, Bangladesh
Embedded Systems Engineering Lead and HoD
 - Developing next-generation electric vehicle (EV) internal systems as an Embedded Systems Engineer, leading teams, and driving efforts to build a custom autonomous car.
- **Adorsho PraniSheba**  Dhaka, Bangladesh
IoT Engineer
 - Developed an AIoT-based wearable device for cow estrus cycle detection, enhancing livestock reproductive efficiency through smart monitoring. (PCB and Firmware)
 - Integrated TinyML, custom raw I²C drivers, BLE beacons, and MQTT to enable real-time, low-power, wireless communication and edge intelligence.
 - Performed comprehensive analysis of low-power system design and sensor signal data to optimize performance for continuous field deployment.
- **FronTech Ltd.**  Sr. Executive (R&D and Production)
 - Engineered a high-performance IIoT and AIoT system for predictive maintenance and advanced industrial control applications.
 - Developed innovative methods for real-time power monitoring, active power factor correction (PFC), and predictive fault detection.
 - Integrated FFT-based signal analysis, Edge Computing, TinyML, and MQTT protocol with custom API solutions for efficient data processing and remote control.
- **Neways International**  IoT Engineer
 - Designed and developed an Intelligent Home Automation System with Integrated Smart Security.

- Engineered a robust fingerprint authentication system featuring Global Fingerprint Sync for seamless and reliable access control across multiple devices.
- Implemented 1:N fingerprint matching using MicroPython and custom API integration, ensuring high-speed identification and enhanced security.
- **ZeronTech BD** 
Part-Time Assistant IoT Engineer
 - IoT Product Development and Inspection.

EDUCATION

- **University of Dhaka** 2018 – 2022
B.Sc. in Electrical and Electronic Engineering Dhaka, Bangladesh
 - **Relevant Coursework:** Digital Signal Processing, Microprocessors, Power Electronics, Control Systems, C++

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [J.1] MD Tanvir Shakil, et al. (2023). **Designing and Implementation of Low-Cost DTMF Based Remotely Controlled Irrigation System with Interactive Voice Response in Bangladesh.** In *International Journal of Science and Business*, DOI: <https://doi.org/10.5281/zenodo.7534982>
- [J.2] MD Tanvir Shakil, et al. (2025). **An Innovative Embedded Ventilator for Accessible and Intelligent Respiratory Support.** In *Journal of Medical and Health Studies*, DOI: <https://doi.org/10.32996/jmhs.2025.6.1.14>
- [J.3] MD Tanvir Shakil, et al. (2025). **AI-Enhanced Multifunctional Smart Assistive Stick for Enhanced Mobility and Safety of the Visually Impaired.** In *Journal of Computer Science and Technology Studies*, DOI: <https://doi.org/10.32996/jcsts.2025.7.1.21>
- [C.1] MD Tanvir Shakil, et al. (2025). **Design and Implementation of an IoT Based Energy-Efficient User-Friendly Smart Home Automation System for Energy Savings and Maximizing Comfort.** In *International Conference on Electrical, Computer and Communication Engineering (ECCE)*, DOI: 10.1109/ECCE64574.2025.11013781.

SKILLS

- **Programming Languages:** C, C++, Python, Embedded C
- **PCB Design Tools:** Altium Designer, EasyEDA
- **Microcontroller and OBC:** STM32, ESP32, AVR, MSP, Raspberry Pi, Jetson Nano
- **Web Technologies:** HTML, JavaScript, REST API, MQTT, CoAP
- **Data Science & Machine Learning:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib
- **Cloud Technologies:** AWS IoT Core, Google Cloud IoT, Firebase
- **DevOps & Version Control:** Git, GitHub
- **Specialized Area:** Embedded Systems, IIoT/IoT/AIoT, RTOS
- **Mathematical & Statistical Tools:** MATLAB, SciPy, R
- **Research Skills:** System Design, Experimental Setup, Simulation, Data Analysis, Technical Writing, Literature Review

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Professional Proficiency), Hindi (Conversational)

Interests: Embedded Systems Design, Robotics, IoT-based Smart Agriculture

REFERENCES

1. **Dr. Mohammad Junaebur Rashid**
Professor, Dept. of Electrical and Electronic Engineering
University of Dhaka, Bangladesh.
Email: mjrashid@du.ac.bd
Relationship: Teacher
2. **Muaz Rahman**
Assistant Professor and Department Head, Department of Electrical and Electronic Engineering (EEE)
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Relationship: Teacher